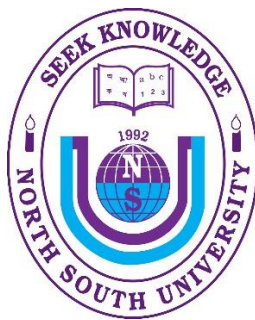


ETE 498R

**Beyond Three Dimensions: An Intuitive Geometric Framework for
Linear Optimization in Four Variables**

This thesis-based project has been submitted to the Department of Electrical and Computer Engineering as the partial fulfillment of the requirements of the Bachelor of Science in Electrical and Telecommunication Engineering (ETE)



Department of ECE
School of Engineering and Physical Sciences (SEPS)
North South University

Declaration

- 1) I hereby declare that my thesis is an original piece of work.
- 2) I declare that, to my knowledge, no part of this work has been published anywhere before.
- 3) I also declare that after publication in a peer-reviewed/ indexed journal, my research work will be considered open to the public.

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Approval

We have reviewed the content of this thesis and are satisfied with its authenticity; therefore, we recommend its approval.

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Abstract

The majority of work done in linear optimization is based on algebraic methods, which propose cumbersome calculations and unintelligible approaches to problems that seem naturally instinctive to human beings and culture. And of course, a general sense of intuition and simplicity are lost within this rich set of mathematical methods and operations, when the number of decision variables rises from sets of two or three. This thesis work investigates the potential usability of geometric/graph-based linear optimization, which is the most intuitive and straightforward solution to optimization-based problems, particularly linear maximization, when working with more than three decision variables. The proposed model provides a visual and insightful representation of working with complex linear models and investigates itself, specifically on the question of whether it outperforms the earlier mentioned algebraic methods when applied to practical implementations of real-life optimization problems.

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Chapter 1

Introduction

1.1 Historical Background

“The problem of deciphering a system of linear inequalities dates back to at least as far as Fourier, who in 1827 published a method for solving them” [12]. The “Fourier-Motzkin elimination” method, also known as the FME method for eliminating variables from a system of linear inequalities, stems from his endeavors in this field. After this, “in 1939, a soviet economist Leonid Kantorovich proposed a method for solving a problem similar to the general linear programming problem” [10]. “He developed an approach to plan expenditures and returns to reduce the cost of the USSR army and increase losses suffered by the communist nation’s adversaries, although this work was initially neglected in the USSR” [3].

“About the same time, A Dutch-American economist, T.C. Koopmans, formulated classical economic problems as linear programs; Kantorovich and Koopmans later shared the 1975 Nobel prize in economics” [12]. “During the time period 1946-1947, George B. Dantzig independently developed the general linear programming formulation method and the simplex method, which for the first time efficiently tackled the linear programming problem in most cases for the US Air Force” [2]. “A formal proof was provided shortly after in an unpublished report, “A Theorem on Linear Inequalities,” on January 5, 1948” [2].

1.2 Significance

Linear Programming is a mathematical method for determining optimal scenarios given any number of linear inequalities, with a diverse range of applications, predominantly in mathematics and operational research, but also in other areas such as business, economics, and numerous engineering applications. Furthermore, many industries rely on this optimization practice, including transportation, energy, telecommunications, and manufacturing, among others.

“The theory behind linear programming drastically reduces the number of possible solutions that must be checked, resulting in drastic computational advantages for its operators. Many of the problems encountered in the real world involve the maximization or minimization of specific quantities. Frequently, these quantities are profit and cost. Linear programming is a collection of procedures for maximizing or minimizing linear functions subject to given linear constraints, much as the functions to be optimized are linear” [14].

Thus, we can see that linear programming, with its vast number of procedures, excels at solving several real-life problems for which other methods, such as calculus, might not render a specific

and a concrete result. Its study, therefore, is paramount to managerial or organizational tasks. Aided by the modern era's computational power, linear programming can organize our lives better than we could achieve iteratively. One last discrepancy to note is that, in the term “Linear Programming”, “programming” has nothing to do with computer programs, as much as it has to do with planning [14].

1.3 Research Topic

As stated in the previous segment, linear maximization, a branch of linear programming, employs various procedures to conduct its operations. Some of them are iterative, while others are highly visual and intuitive. The most widely used methods are as listed in the next page.

Table 1.1: Operation Procedures and Their Descriptions

Procedure Name	Description
Simplex	“Simplex Method is an iterative procedure allowing the improvement of the solution by each step, it is very effective for canonical maximizations and minimizations”. [14] Steve Smale [13] Wrote: “The simplex method of George Dantzig is extensively used to solve linear programming problems and is extremely efficient.”
Dual Simplex	“The Dual Simplex transforms canonical maximization problems into canonical minimization problems by negative transposition” [14]. Once this task is accomplished, we continue with the normal simplex method.
Big [M]	“The Big [M] or penalty method is based on the simplex method. This method however introduces artificial variables in the objective function to get an initial basic feasible solution (IBFS). Artificial variables are then removed using iterative process which also leads to an optimum solution” [7].
Geometric Approach	The geometric or graphical method of linear programming is an intuitive and visual method that does not rely on continuous iterations. However, it is only limited to Two or Three Decision Variables [8].

In all the procedures available in linear programming, ergo, linear maximization we can see that the Geometric approach is the most Intuitive and friendly to people first approaching the subject, but it is deemed unusable when complex canonical optimizations with more than three decision variables need to be operated on [8]. We have tested the merit of this claim in this paper, the result of which will be illustrated upon elaborately.

1.4 Motivation

Most real-life problems have numerous constraints, but most of these constraints are weak constraints, so they can be generally ignored during modelling the linear program.

Typically, most problems can be modelled into problems containing three decision variables; these problems are then solved through iterative approaches, the calculations of which are cumbersome [16]. This requires a significant amount of computational power if our model contains more than three decision variables.

Thus, we sought to determine if linear optimization can be applied to more than three decision variables using the geometric approach. In this effort, we attempt to solve a set of inequalities with more than three decision variables in this paper using the geometric approach of linear optimization.

1.5 Objectives

- Formulating a linear model with four decision variables
- Plotting constraints on a four-dimensional graph
- Find the solution space for the introduced model
- Find the optimum solution for linear maximization using our method
- Comparing our solution with that of the Google GLOP LP solver
- Describe the accuracy of our method

1.6 Contribution Summary

- A novel approach to solving models with four decision variables.
- Procedure of working through a linear program in the fourth dimension through a hyper-tetrahedral.
- Introduction of solution spaces, which are used as intersection detectors for higher dimensions.

1.7 Thesis Outline

The total report is organized as follows:

- Chapter 2 contains a summary of the available methods for geometric maximizations.
- Chapter 3 contains an in-depth discussion and explanation of our method.
- Chapter 4 contains results and findings.
- Chapter 5 is a discussion about the impacts and ethics of work and the challenges we faced.
- Chapter 6 contains probable alternatives to the method we proposed and the constraints that we had to operate around.
- Chapter 7 is a visual representation of the whole work's timeline and milestones, along with some discussion on how it was implemented.
- Chapter 8 concludes our thesis work and suggests future improvements.

Chapter 2

Background

2.1 Literature Review

Most linear optimization problems are addressed using algebraic methods [1][2]. However, when most literatures on this topic is studied an amusing commonality can be found within them, they all tend to introduce the subject using a relatively simple mechanic called the geometric/graphical approach to solving linear programs [2][4] [8][10][12][15][16], But then they move on to the algebraic methods saying the visualization for three or more decision variables can be rather complex and from four decision variables, it is regarded impossible to be operated on by the graphical approach [8].

This happens because of the special dimension that must be graphed and meticulously measured to find the answer to each problem. So, the algebraic methods are consequently introduced, and for the remainder of one's career in operational research, they would rather choose the reliable algebraic methods [1][2] to deal with optimization-based problems. Some attempts at utilization the graphical method is implemented when working with three decision variables or optimization problems where the decision variables can be reduced until only three is left [5][6], but when these are reviewed, the complexity with which they have been operated upon seems to overwhelm any who would dare seek to utilize the graphical method to solve any real life optimization problems intuitively.

However, upon reviewing the mathematical complexity that the algebraic methods produce in their implementation [7][9][13], it seems more reasonable to utilize something simpler and more relatable to humans.

2.2 Model Formulation

Linear programming, also known as linear maximization, begins with model formulation. Formulation is the act of preparing something. Given a real-life problem, we first try to organize a model of linear inequalities that we can solve using the procedures that linear programming makes available for us to utilize. Let us then examine a sample problem to understand how the graph model of linear programming works.

Table 2.1: Sample Resource Data

Resource Data			
Product	Labor	Material Cost	Profit
Pen	1	4	4
Pencil	2	3	5

Given that, profit = Sales – Total Cost.

And Total Cost = Labor + Material Cost.

Here, we can see that we have two decision variables. Let X and Y denote each of them,

X = Number of pens produced per day.

And, Y = Number of pencils produced per day.

Our objective being, the maximization of the profit. Let Z denote the profit.

Thus, our objective function is, Maximize $Z = 4X + 5Y$.

We also have two constraint functions in our formulated model, constrained functions are the limitations put on our production system.

Supposing, we cannot have more than 4 labors working for us at any given time and the material cost cannot exceed 12, our constraint functions will look something like...

$$1X + 2Y \leq 4$$

And, $3X + 4Y \leq 12$

We also, in most cases harbor a non-negativity constraint, which can be denoted as...

$$X, Y \geq 0$$

Let us now look at our completed linear model,

$$\text{Maximize } Z = 4X + 5Y$$

$$\text{Subject to, } 1X + 2Y \leq 4$$

$$3X + 4Y \leq 12$$

$$\& X, Y \geq 0$$

2.3 Solving proposed Sample Model

To solve the proposed linear model, we need to find the optimum feasible solution. Feasible solutions are essentially solutions that does not violate any of the constraint functions [8]. To

find these feasible solutions, we have to plot a n-dimensional graph of the constraint functions and find a feasible region, “n” being the number of decision variables [4].

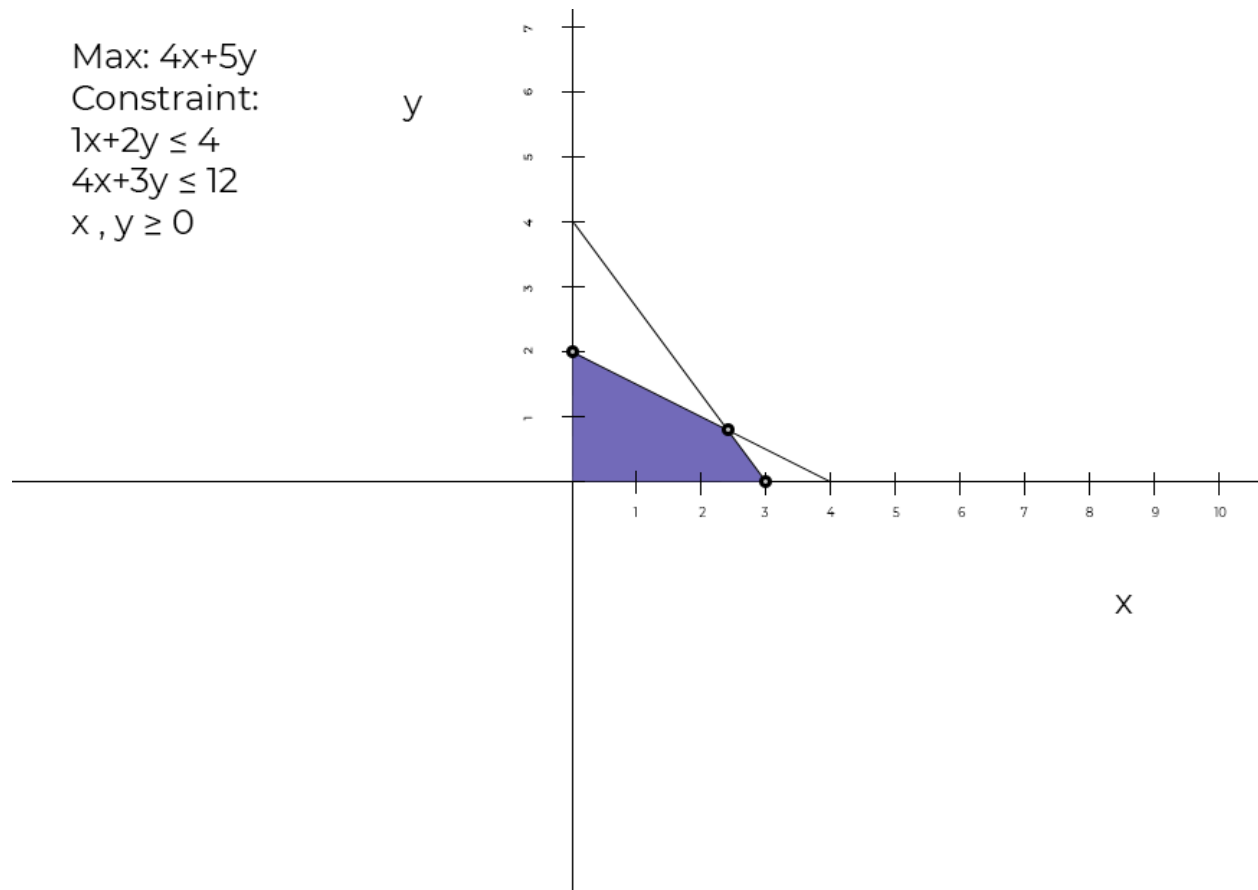


Figure 2.1: Feasible Region of Sample Linear model

Since, the condition to find the feasible solution is that they cannot violate any of the constraint functions, they have to be contained in an area/simplex where both constraint functions are valid. This simplex is made of the intersection points of the constraint functions after being plotted on the n-dimensional graph [16]. This simplex is called the “Feasible Region” and the intersecting constraint points that construct this geometric shape are the “Feasible Solutions”

Since, we are maximizing our objective function, our objective is to find a workable solution where the objective function is the greatest. In the case of this sample linear model this feasible solution can be found at

$$X = 2.4 \text{ and } Y = 0.8$$

And our Objective function renders the value of 13.6.

This can be further backed up when compared to the answers given by the “Google Glop Linear Solver”, which uses the simplex method to solve Linear models, when tested for the same linear Model.

```
☞ Number of variables = 2  
   Number of constraints = 2  
   Solution:  
   x = 2.4  
   y = 0.8  
   Optimal objective value = 13.6
```

Figure 2.2: GLOP Linear Solver Solution to Sample Linear Model

Chapter 3

Proposed Method

3.1 General Observations and Concept Introduction

Before we delve into the proposed method, we must first list out some general observations and introduce some concepts which will become necessary in our pursuit of solving linear models containing more than the proposed amount of decision variables that the vanilla graphical method limits itself to [8].

Binary Solution Space: The binary solution space is a fictional table containing a number of solution spaces. It has 2^{n+1} columns and $n+2$ rows, which in itself includes every possible intersection that can occur in that n -dimensional graph, given that n is the number of decision variables. The rows contain each of the solution spaces.

We can then compare the Binary solution space to the feasible region of a linear model and the rows to the possible solutions.

From 1 to n , the columns contain each decision variable, the last two columns are reserved for the results yielded by the optimized objective function when tested for the specific solution space and the Object set of the particular solution space.

Object Set: Given that every constraint function projects an object when plotted in a graph, be it a triangle or a tetrahedron, the object set is the set of objects that reside in a particular solution space that might or might not intersect with each other to create said solution space.

For example, for the vanilla 2-dimensional graph containing 2 decision variables, we construct the binary solution space as follows.

Number of Decision Variables: 2

Table 3.1: Sample 2-Dimensional Binary Solution Space

X	Y	Objective Function	Object Set
0	0	Some Value	{some set of objects}
1	0	Some Value	{some set of objects}
0	1	Some Value	{some set of objects}
1	1	Some Value	{some set of objects}

Where 0 represents the possibility of there being no feasible solution existing for a specific plane in a particular space of solution, and 1 represents the possibility of there being a feasible solution.

If carefully observed, we can see that possibility values in the solution spaces are organized as if one were traversing the possible binary values that can be found in 2 bits. Hence, the naming of the binary solution space.

Supposing we have three constraint variables which create three tetrahedra when plotted in a graph, let us make another ample binary solution space:

Table 3.2 Sample 3-Dimensional Binary Solution Space

X	Y	Z	Objection Function	Object Set
0	0	0	Some Value	{some set of objects}
1	0	0	Some Value	{some set of objects}
0	1	0	Some Value	{some set of objects}
0	0	1	Some Value	{some set of objects}
1	1	0	Some Value	{some set of objects}
1	0	1	Some Value	{some set of objects}
0	1	1	Some Value	{some set of objects}
1	1	1	Some Value	{some set of objects}

Observations:

- When maximizing, in general cases, the object or the intersection point with the lowest plotted value in any plane of an n-dimensional graph forms the feasible region simplex.
- When minimizing, in general cases, the object or the intersection point with the highest plotted value in any plane of an n-dimensional graph forms the feasible region simplex.
- Intersection only happens between multiple objects when the lines formed in each plane by every object aren't parallel to each other. Thus, given an XY plane and two triangles, they will only intersect if each of the edges is plotted on planar values that cannot be divided by a common integer. An example of two triangles that won't intersect are {2,4,6} and {4,6,8}, on the contrary two triangles that will intersect are {2,4,6} and {4,5,8}
- In the feasible region, we can ignore all other intersections apart from the one closest to the solution simplex.

3.2 Algorithm

Given that the concepts mentioned above have been appropriately understood, here's our general algorithm for solving any linear model in the geometric procedure.

1. Formulate the objective and constraint functions.
2. From the constraint function, extrude the geometric shapes and mark each constraint-derived shape as its own object.
3. "n" being the number of decision variables, for each object, plot them on the n-dimensional graph.
4. Construct the binary solution space.
5. Refine the binary solution space based on the feasible solutions, writing out the objects involved in the intersected point in the object set.
6. Solve the refined binary solution space utilizing the elimination method on constraints based on the object set.
7. Compare and find the optimized solution.

3.3 Solving Poly-dimensional Linear Models Using the Proposed Method

For the linear model containing 3 Decision variables:

Maximize Z: $x + 5y + 5z$

Subject to: $4x + 6y + 3z \leq 24$

$$x + 2y + 3z \leq 12$$

$$3x + 2y + z \geq 6$$

$$\& x, y, z \geq 0$$

Step 1:

Objective Function: $Z = x + 5y + 5z$

Constraint Functions: $4x + 6y + 3z \leq 24$

$$x + 2y + 3z \leq 12$$

$$3x + 2y + z \geq 6$$

$$\& x, y, z \geq 0$$

Step 2:

Tetrahedron Object 1: $\{6, 4, 8\}$

Tetrahedron Object 2: $\{12, 6, 4\}$

Tetrahedron Object 3: $\{2, 3, 6\}$

Step 3:

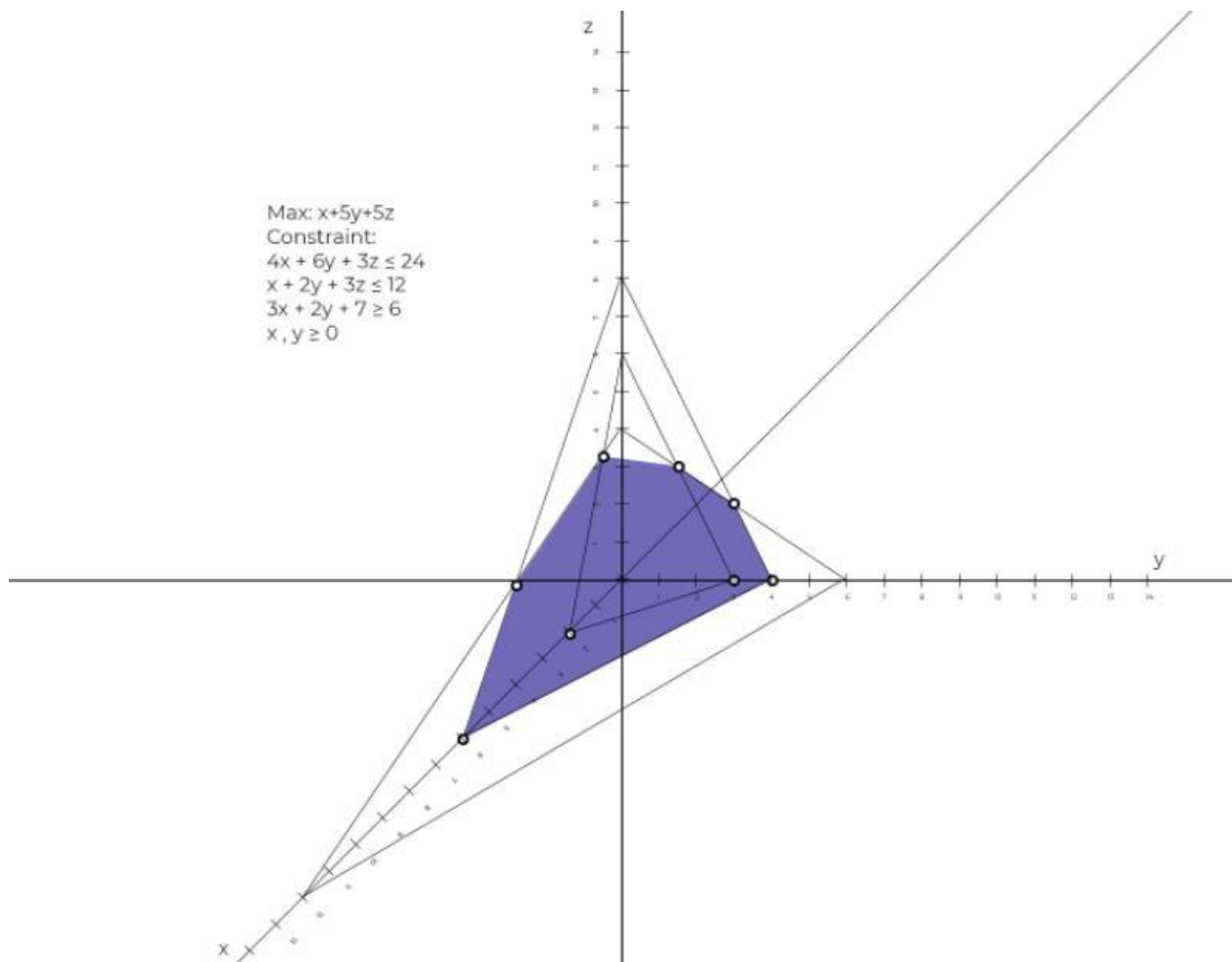


Figure 3.1: 3-Dimensional Graph

Step 4:

Table 3.3: Binary Solution Space for 3-Dimensional Model

X	Y	Z	Objection Function	Object Set
0	0	0	Some Value	{some set of objects}
1	0	0	Some Value	{some set of objects}
0	1	0	Some Value	{some set of objects}
0	0	1	Some Value	{some set of objects}
1	1	0	Some Value	{some set of objects}

1	0	1	Some Value	{some set of objects}
0	1	1	Some Value	{some set of objects}
1	1	1	Some Value	{some set of objects}

Step 5:

Table 3.4: Refined Binary Solution Space for 3-Dimensional Model

X	Y	Z	Objection Function	Object Set
1	0	0	Some Value	{1}
1	0	0	Some Value	{3}
0	1	0	Some Value	{1}
0	1	0	Some Value	{3}
1	0	1	Some Value	{1,2}
1	0	1	Some Value	{2,3}
0	1	1	Some Value	{1,2}
0	1	1	Some Value	{3,2}

Step 6:

Table 3.5: Solved refined Binary Solution Space for 3-Dimensional Model

X	Y	Z	Objection Function	Object Set
1	0	0	6	{1}
1	0	0	2	{3}
0	1	0	20	{1}
0	1	0	12	{3}
1	0	1	17.33	{1,2}
1	0	1	19.5	{2,3}
0	1	1	25	{1,2}
0	1	1	22.5	{3,2}

Step 7:

We can therefore see that, for this specific linear model the solution lies in the 7th Solution Space of the refined binary solution space, in which object 1 and 2 intersect with each other in the YZ plane, in the co-ordinate {0,3,2}

And the optimized Objective function yields a value of $Z = 25$

Using our algorithm, we can see that solving a system of three decision variables is quite intuitive and easy to implement when compared to the system suggested by T.R. Kannan et al [7]. Now let us attempt to solve a linear model containing 4 decision variables.

For the linear model containing 4 Decision Variables:

Maximize S: $20W + 18X + 3Y + 10Z$

Subject to: $15W + 15X + 3.75Y + 3Z \leq 30$

$$5W + 8X + 4Y + 5Z \leq 40$$

$$6W + X + 2Y + 5Z \leq 10$$

$$\& w, x, y, z \geq 0$$

Step 1:

Objective Function: $S = 20W + 18X + 3Y + 10Z$

Constraint Functions: $15W + 15X + 3.75Y + 3Z \leq 30$

$$5W + 8X + 4Y + 5Z \leq 40$$

$$6W + X + 2Y + 5Z \leq 10$$

$$\& w, x, y, z \geq 0$$

Step 2:

Hyper-Tetrahedron Object 1: $\{8, 5, 10, 8\}$

Hyper-Tetrahedron Object 2: $\{2, 2, 8, 10\}$

Hyper-Tetrahedron Object 3: $\{1.6, 10, 5, 2\}$

Step 3:

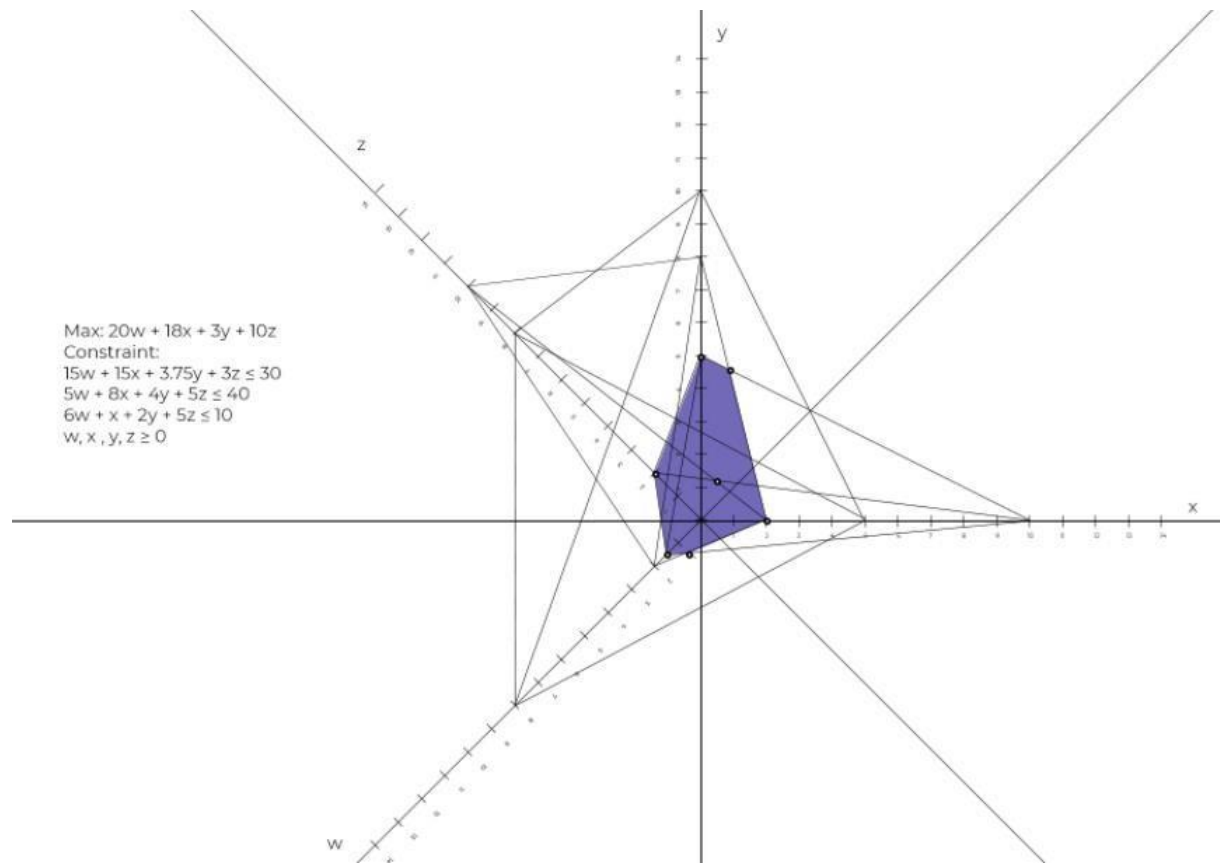


Figure 3.2: 4-Dimensional Graph

Step 4:

Table 3.6: Binary Solution Space for 4-Dimensional Model

W	X	Y	Z	Objection Function	Object Set
0	0	0	0	Some Value	{some set of objects}
1	0	0	0	Some Value	{some set of objects}
0	1	0	0	Some Value	{some set of objects}
0	0	1	0	Some Value	{some set of objects}
0	0	0	1	Some Value	{some set of objects}
1	1	0	0	Some Value	{some set of objects}

1	0	1	0	Some Value	{some set of objects}
1	0	0	1	Some Value	{some set of objects}
0	1	1	0	Some Value	{some set of objects}
0	1	0	1	Some Value	{some set of objects}
0	0	1	1	Some Value	{some set of objects}
1	1	1	0	Some Value	{some set of objects}
1	0	1	1	Some Value	{some set of objects}
1	1	0	1	Some Value	{some set of objects}
0	1	1	1	Some Value	{some set of objects}
1	1	1	1	Some Value	{some set of objects}

Step 5:

Table 3.7: Refined Binary Solution Space for 4-Dimensional Model

W	X	Y	Z	Objection Function	Object Set
1	0	0	0	Some Value	{3}
0	1	0	0	Some Value	{2}
0	0	1	0	Some Value	{3}
0	0	0	1	Some Value	{3}
1	1	0	0	Some Value	{2,3}
0	1	1	0	Some Value	{2,3}
0	1	0	1	Some Value	{2,3}

Step 6:

Table 3.8: Solved refined Binary Solution Space for 4-Dimensional Model

W	X	Y	Z	Objection Function	Object Set
1	0	0	0	32	{3}
0	1	0	0	36	{2}
0	0	1	0	15	{3}
0	0	0	1	20	{3}
1	1	0	0	39.2	{2,3}

0	1	1	0	27.9	{2,3}
0	1	0	1	46.6	{2,3}

Step 7:

We can therefore see that, for this specific linear model, the solution lies in the 7th Solution Space of the refined binary solution space, in which hyper-tetrahedraon 2 and 3 intersect with each other in the XZ plane, in the co-ordinate {0,1.6,0,1.6}

And the optimized Objective function yields a value of $S = 46.6$

Thus, using this model we have solved a linear program model with four decision variables

Chapter 4

Result Analysis

4.1 Result Comparison

Using the Google Glop linear solver, we have solved the aforementioned linear models and numerous others, and then compared the results with our method, the results of which will be discussed in this chapter.

For the linear model

Maximize Z: $x + 5y + 5z$

Subject to: $4x + 6y + 3z \leq 24$

$$x + 2y + 3z \leq 12$$

$$3x + 2y + z \geq 6$$

$$\& x, y, z \geq 0$$

Our answer, as illustrated in Chapter 3, was 25.

And the Google GLOP LP solver's answer was:

```

☐➤ Number of variables = 3
    Number of constraints = 3
    Solution:
    x = 0.0
    y = 3.0000000000000004
    z = 1.9999999999999996
    Optimal objective value = 25.0

```

Figure 4.1 GLOP answer for problem 1

For the linear model

Maximize S: $20W + 18X + 3Y + 10Z$

Subject to: $15W + 15X + 3.75Y + 3Z \leq 30$

$$5W + 8X + 4Y + 5Z \leq 40$$

$$6W + X + 2Y + 5Z \leq 10$$

$$\& w, x, y, z \geq 0$$

Our answer, as illustrated in Chapter 3, was 46.6.

And the Google GLOP LP solver's answer was:

```

C> Number of variables = 4
    Number of constraints = 3
    Solution:
    w = 0.0
    x = 1.6666666666666663
    y = 0.0
    z = 1.6666666666666665
    Optimal objective value = 46.66666666666666
  
```

Figure 4.2 GLOP answer for problem 2

For the linear model

Maximize S: $20W + 18X + 3Y + 10Z$

Subject to: $5W + 15X + 3Y + 4Z \leq 30$

$5W + 8X + 4Y + 5Z \leq 40$

& $w, x, y, z \geq 0$

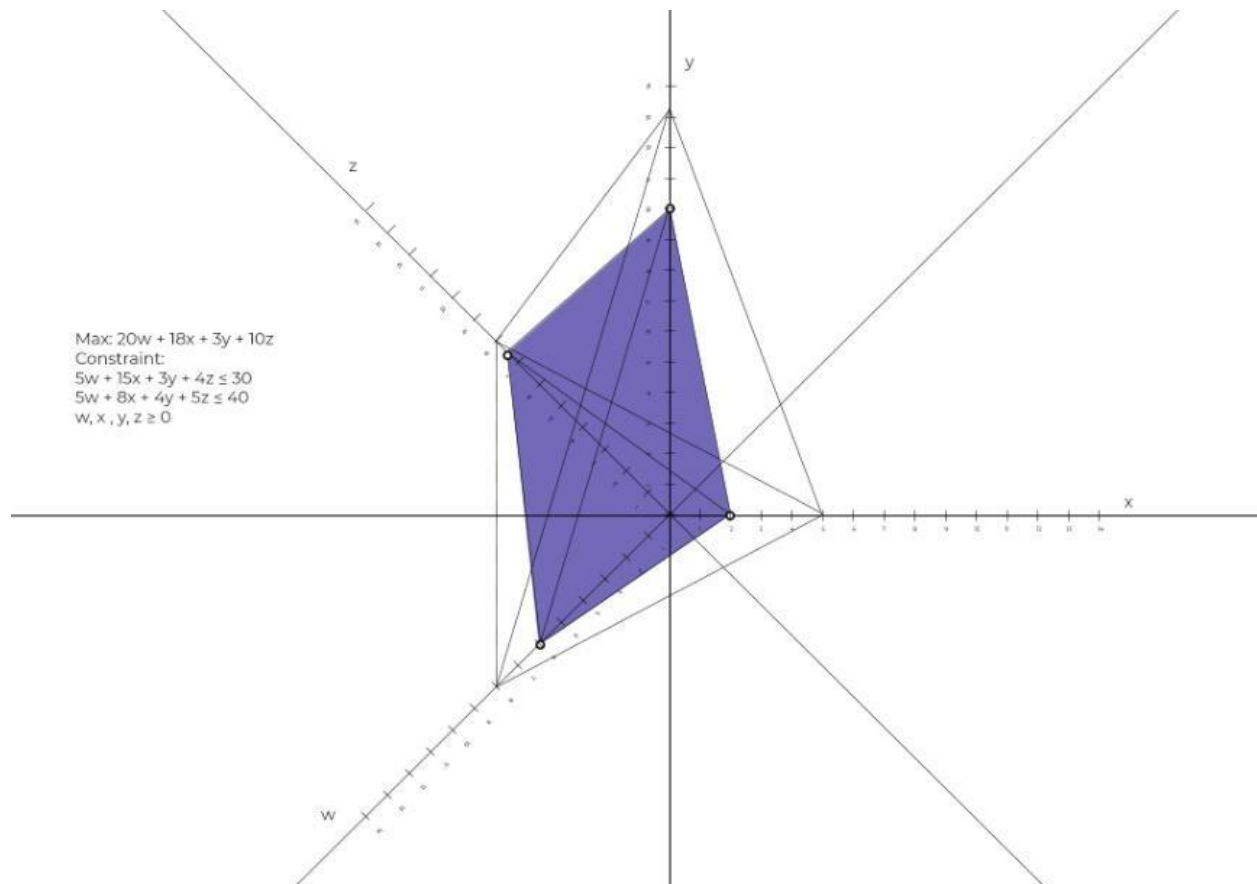


Figure 4.3: 4-Dimensional Graph for Problem 3.

Our answer was 120 in the W {6,0,0,0} plane for object 1

And the Google GLOP LP solver's answer was:

```

↳ Number of variables = 4
   Number of constraints = 2
   Solution:
   w = 6.0
   x = 0.0
   y = 0.0
   z = 0.0
   Optimal objective value = 120.0

```

Figure 4.4 GLOP answer for problem 3

For the linear model

Maximize S: $20W + 18X + 3Y + 10Z$

Subject to: $5W + 15X + 3.75Y + 3Z \leq 30$

$5W + 8X + 4Y + 5Z \leq 40$

& $w, x, y, z \geq 0$

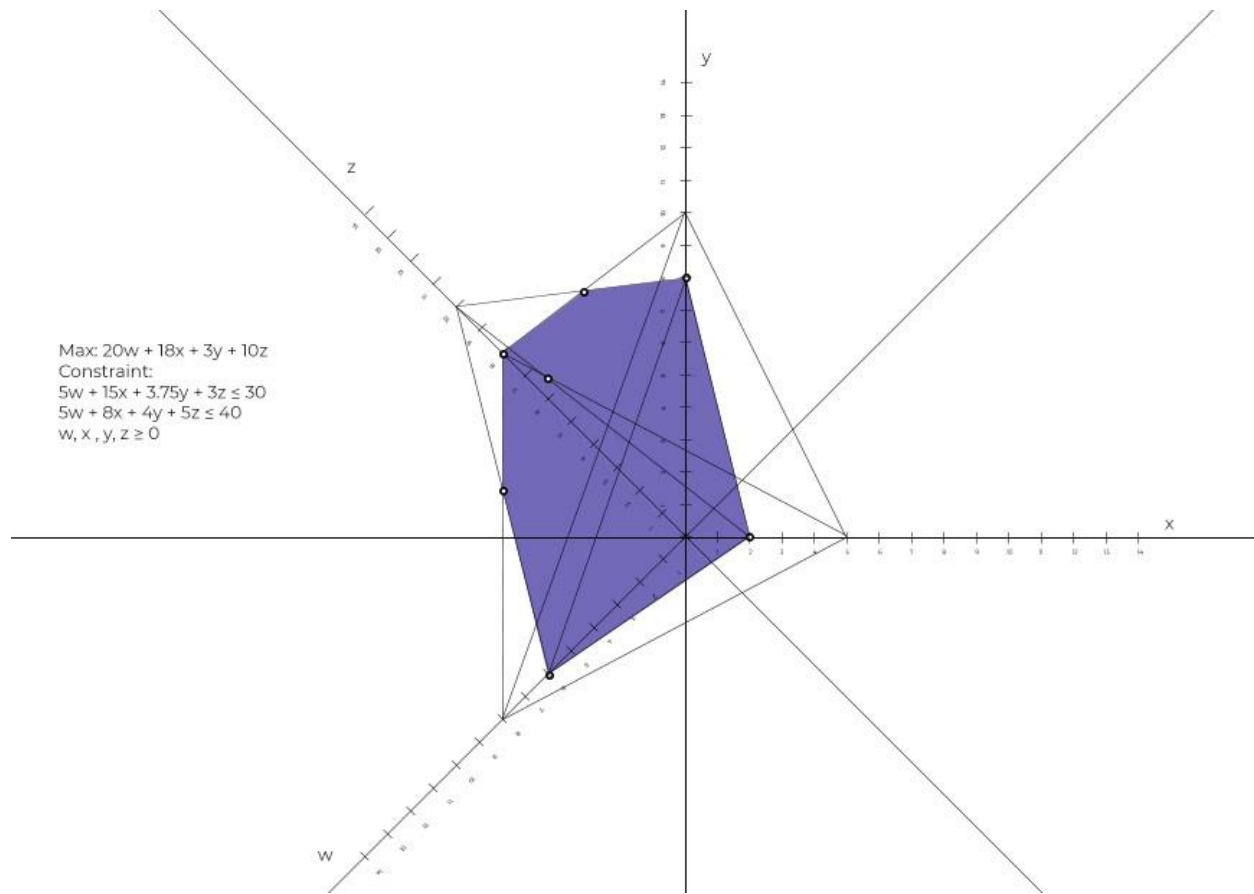


Fig 4.5 4-Dimensional Graph for Problem 4.

Our answer was 120 in the W {6,0,0,0} plane for object 1

And the Google GLOP LP solver's answer was:

```

↳ Number of variables = 4
   Number of constraints = 2
   Solution:
   w = 6.0
   x = 0.0
   y = 0.0
   z = 0.0
   Optimal objective value = 120.0

```

Figure 4.6 GLOP answer for problem 4

And lastly,

For the linear model

Maximize S: $20W + 18X + 3Y + 10Z$

Subject to: $15W + 15X + 3.75Y + 3Z \leq 30$

$$5W + 8X + 4Y + 5Z \leq 40$$

$$6W + X + 2Y + 5Z \leq 10$$

$$10W + 10X + 10Y + 10Z \geq 10$$

$$\& w, x, y, z \geq 0$$

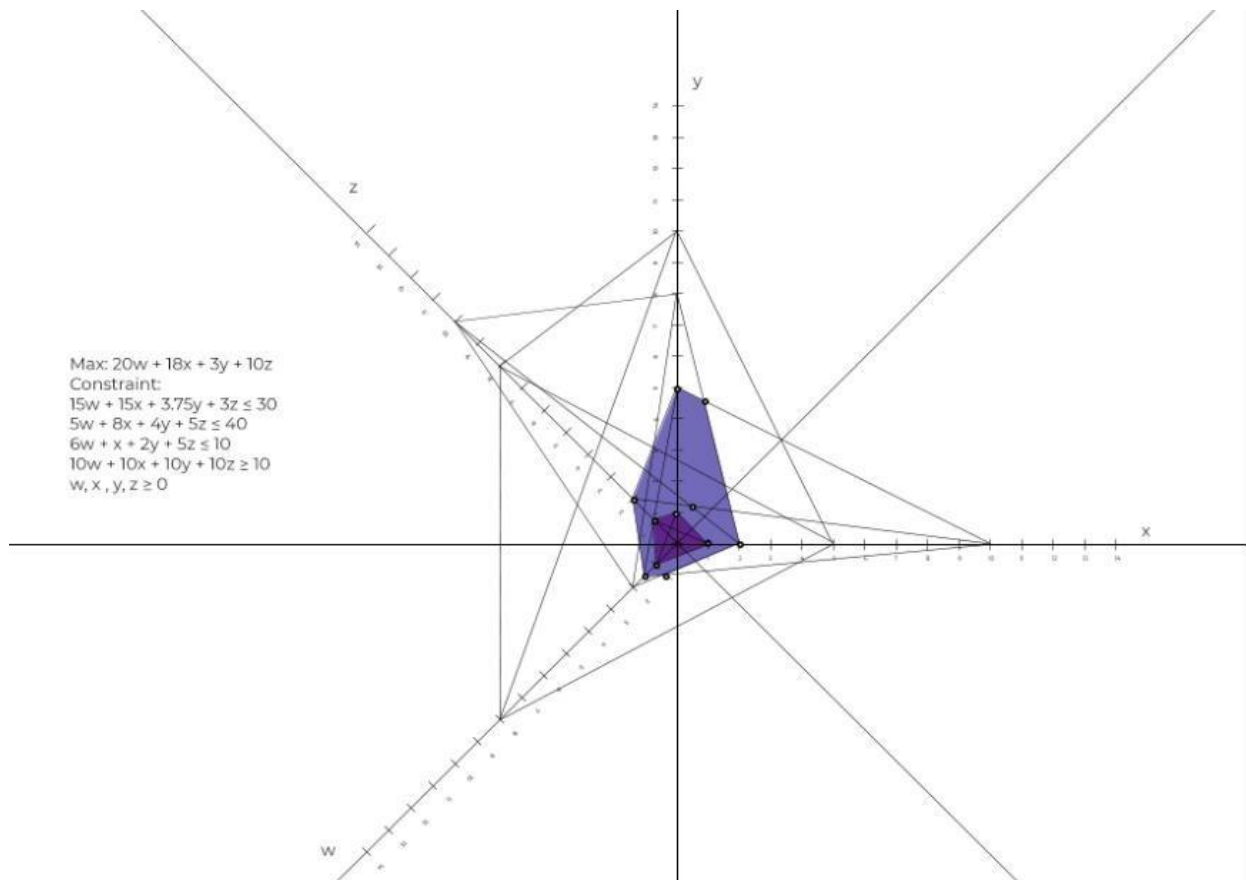


Fig 4.7 4-Dimensional Graph for Problem 5

Our answer was 46.6. The Feasible solution is at the XZ plane at $\{0, 1.6, 0, 1.6\}$ in the intersection between hyper-tetrahedron object 2 and hyper-tetrahedron object 3

And the Google GLOP LP solver's answer was:

```

Number of variables = 4
Number of constraints = 4
Solution:
w = 0.0
x = 1.6666666666666663
y = 0.0
z = 1.6666666666666667
Optimal objective value = 46.666666666666666

```

Figure 4.8 GLOP answer for problem 5

Similarly, we have solved all of these proposed models and compared them to Google's linear solver:

Table 4.1: Mass result comparison between the conventional linear solver and our method

Proposed Linear Model	Our Answer	Google GLOP Answer
Maximize $S = 20W + 18X + 3Y + 10Z$ subject to $15W + 10X + 3.75Y + 3Z \leq 30$ $5W + 8X + 4Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $10W + 10X + 10Y + 10Z \geq 10$	60.85	60.8511
Maximize $S = 4W + 5X + 3Y + Z$ subject to $15W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $10W + 3X + 10Y + Z \geq 10$	15	15
Maximize $S = 40W + 5X + 3Y + Z$ subject to $15W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $W + 3X + 10Y + Z \geq 10$	51.4	51.4
Maximize $S = W + 15X + 13Y + 10Z$ subject to $5W + 10X + 6Y + 3Z \leq 30$ $6W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 3X + 10Y + Z \geq 10$	65	65
Maximize $S = 4W + 3X + 3Y + 18Z$ subject to $15W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $10W + 3X + 10Y + Z \geq 10$	34.1	34.0625
Maximize $S = 2W + 5X + 13Y + 10Z$ subject to $5W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 3X + 10Y + Z \geq 10$	65	65
Maximize $S = 2W + 5X + Y + Z$ subject to $5W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 3X + 10Y + Z \geq 10$	14.76	14.7561

Maximize $S = 4W + 15X + 3Y + 18Z$ subject to $15W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $10W + 3X + 10Y + Z \geq 10$	64.1	64.0625
Maximize $S = 4W + 18X + 3Y + 10Z$ subject to $15W + 10X + 6Y + 3Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 10$ $10W + 3X + 10Y + Z \geq 10$	59.9	59.8958
Maximize $S = 2W + 15X + Y + Z$ subject to $3W + 14X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + Z \geq 10$	32.14	32.1429
Maximize $S = 12W + 15X + Y + Z$ subject to $3W + 14X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + Z \geq 10$	48.5	48.4884
Maximize $S = W + 15X + Y + Z$ subject to $3W + 14X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + Z \geq 10$	32.14	32.1429
Maximize $S = 4W + 5X + 10Y + Z$ subject to $30W + 14X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	50	50
Maximize $S = 4W + 9X + 10Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	57.7	57.6923
Maximize $S = 40W + 9X + 10Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $14W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	111.1	111.111
Maximize $S = 40W + 29X + 10Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$	242	242

$2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$		
Maximize $S = 4W + 2X + Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	20	20
Maximize $S = 7W + 2X + Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	33.64	33.6364
Maximize $S = 7W + 2X + 8Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	40	40
Maximize $S = 5W + 7X + 8Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + 10Y + 5Z \geq 10$	47.2	47.2
Maximize $S = 5W + X + 8Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$ $4W + 6X + 10Y + 5Z \geq 10$	25	25
Maximize $S = W + X + 4Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	14	14
Maximize $S = W + X + 14Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	35	35
Maximize $S = 4W + 4X + 4Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	29.3333	29.3333
Maximize $S = 4W + 14X + 4Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	76	76
Maximize $S = 2W + X + 4Y + 5Z$ subject to	10	10

$3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$		
Maximize $S = 7W + 2X + 8Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	35	35
Maximize $S = 3W + 2X + 3Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	17.3333	17.3333
Maximize $S = 14W + 2X + 3Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	70	70
Maximize $S = W + 2X + 3Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	16.7	16.6667
Maximize $S = W + 2X + 3Y + Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + 5Z \leq 10$	12.6	12.5926
Maximize $S = W + 2X + 3Y + Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	12.15	12.1481
Maximize $S = 7W + 2X + 3Y + Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	35	35
Maximize $S = 6W + 2X + 3Y + Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	30	30
Maximize $S = 5W + 2X + 3Y + Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	25	25
Maximize $S = 5W + 5X + 3Y + Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	36.7	36.6667
Maximize $S = 5W + 4X + 3Y + Z$ subject to	32	32

$3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$		
Maximize $S = 5W + 6X + 7Y + 2Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	41.3	41.3333
Maximize $S = 5W + 6X + 7Y + 2Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$ $2W + 4X + 6Y + 3Z \leq 20$	36.7	36.6667
Maximize $S = 5W + 6X + 7Y + 10Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 4Z \leq 40$ $2W + X + 4Y + Z \leq 10$ $2W + 4X + 6Y + 3Z \leq 20$	66.7	66.6667
Maximize $S = 5W + 6X + 7Y + 2Z$ subject to $3W + 4X + Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + Z \leq 10$	41.3333	41.3333
Maximize $S = 5W + 3X + 7Y + 2Z$ subject to $3W + 4X + Y + 3Z \leq 30$ $W + 8X + 5Y + 24Z \leq 40$ $2W + X + 4Y + 2Z \leq 10$	27.3333	27.3333
Maximize $S = 5W + 7X + 8Y + 7Z$ subject to $3W + 4X + 6Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + Y + 5Z \geq 10$	47.2	47.2
Maximize $S = 5W + 7X + 8Y + 7Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + Y + 5Z \geq 10$	46.2	46.1714
Maximize $S = 5W + 7X + Y + 7Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$ $W + 6X + Y + 5Z \geq 10$	46	46
Maximize $S = 5W + 6X + Y + 7Z$ subject to $3W + 4X + 16Y + 3Z \leq 30$ $W + 8X + 5Y + 5Z \leq 40$ $2W + X + 2Y + 5Z \leq 10$	41.3333	41.3333

W + 6X + Y + 5Z >= 10		
Maximize S = 5W + 6X + Y + 7Z subject to 3W + 4X + 16Y + 3Z <= 30 W + 18X + 5Y + 5Z <= 40 2W + X + 2Y + 5Z <= 10 W + 6X + Y + 5Z >= 10	32	32
Maximize S = 5W + 6X + 7Y + 2Z subject to 3W + 4X + Y + 3Z <= 30 W + 8X + 5Y + 25Z <= 40 2W + X + 4Y + Z <= 10	41.3333	41.3333
Maximize S = 5W + 6X + 7Y + 12Z subject to 3W + 4X + Y + 3Z <= 30 W + 8X + 5Y + 5Z <= 40 2W + X + 4Y + Z <= 10	98.9	98.8889
Maximize S = 5W + 6X + 7Y + 2Z subject to 3W + 4X + Y + 3Z <= 30 W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	25	25
Maximize S = 4W + 6X + 5Y + 2Z subject to 3W + 4X + Y + 3Z <= 30 6W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	20	20
Maximize S = 4W + 6X + 5Y + 2Z subject to 3W + 4X + Y + 3Z <= 30 10W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	18	18
Maximize S = 3W + 16X + 5Y + 2Z subject to 3W + 6X + Y + 3Z <= 30 10W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	18.1	18.0645
Maximize S = 3W + 15X + 5Y + 2Z subject to 3W + 6X + Y + 3Z <= 30 10W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	18	18
Maximize S = 3W + 3X + 5Y + 22Z subject to 3W + 6X + Y + 3Z <= 30 10W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	176	176
Maximize S = 3W + 3X + 5Y + 19Z subject to 3W + 6X + Y + 3Z <= 30 10W + 8X + 5Y + 5Z <= 40 2W + 14X + 4Y + Z <= 10	152	152
Maximize S = 3W + 3X + 5Y + 9Z subject to	72	72

$3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$		
Maximize $S = 3W + 3X + 5Y + 14Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	112	112
Maximize $S = 3W + 3X + 5Y + 11Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	88	88
Maximize $S = 3W + 3X + 5Y + 18Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	144	144
Maximize $S = 3W + 3X + 5Y + 15Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	120	120
Maximize $S = 3W + 3X + 5Y + 17Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	136	136
Maximize $S = 3W + 3X + 5Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	48	48
Maximize $S = 3W + 3X + 15Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	54	54
Maximize $S = 3W + 3X + 13Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	52.7	52.6667
Maximize $S = 3W + 3X + 14Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	53.3333	53.3333
Maximize $S = 3W + 3X + 11Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	51.3333	51.3333
Maximize $S = 3W + 3X + 20Y + 6Z$ subject to	57.3333	57.3333

$3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$		
Maximize $S = 3W + 3X + 22Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	58.7	58.6667
Maximize $S = 3W + 3X + 24Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	60	60
Maximize $S = 3W + 3X + 24Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	192	192
Maximize $S = 3W + 3X + 2Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	48	48
Maximize $S = 3W + 3X + 8Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	64	64
Maximize $S = 3W + 3X + 10Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	80	80
Maximize $S = 3W + 3X + 12Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	96	96
Maximize $S = 3W + 3X + 14Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	112	112
Maximize $S = 3W + 3X + 15Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$	120	120
Maximize $S = 3W + 3X + 16Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	128	128
Maximize $S = 3W + 3X + 17Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	136	136
Maximize $S = 3W + 3X + 18Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	144	144
Maximize $S = 3W + 3X + 19Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	152	152

Maximize $S = 3W + 3X + 20Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	160	160
Maximize $S = 3W + 3X + 5Y + 6Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	60	60
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	200	200
Maximize $S = 3W + 3X + 2Y + 19Z$ subject to $3W + X + Y + 3Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	190	190
Maximize $S = 3W + 3X + 2Y + 19Z$ subject to $3W + X + Y + 4Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	142	142
Maximize $S = 3W + 3X + 2Y + 19Z$ subject to $3W + X + Y + 5Z \leq 30$ $4W + 8X + 5Y + 4Z \leq 40$	114	114
Maximize $S = 3W + 3X + 2Y + 19Z$ subject to $3W + X + Y + 6Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$	95	95
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 6Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$	100	100
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 7Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$	86.1	86.0784
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 8Z \leq 30$ $4W + 8X + 5Y + 5Z \leq 40$	76.44	76.4407
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 8Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$	75.71	75.7143
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 9Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$	68.125	68.125
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 10Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$	62.2222	62.2222
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 11Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$	57.5	57.5
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to	53.64	53.6364

$3W + X + Y + 12Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$		
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 13Z \leq 30$ $4W + 8X + 5Y + 8Z \leq 40$	50.42	50.4167
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 13Z \leq 30$ $4W + 8X + 5Y + 10Z \leq 40$	49.57	49.5745
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 13Z \leq 30$ $4W + 8X + 5Y + 11Z \leq 40$	49.12	49.1398
Maximize $S = 3W + 3X + 2Y + 20Z$ subject to $3W + X + Y + 13Z \leq 30$ $4W + 8X + 5Y + 13Z \leq 40$	48.24	48.2418
Maximize $S = 3W + 3X + 2Y + 19Z$ subject to $3W + X + Y + 13Z \leq 30$ $4W + 8X + 5Y + 13Z \leq 40$	46.044	46.044
Maximize $S = 9W + 3X + 24Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	60	60
Maximize $S = 9W + 4X + 23Y + 6Z$ subject to $3W + 6X + Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 14X + 4Y + Z \leq 10$	59.3333	59.3333
Maximize $S = 8W + 4X + 2Y + 6Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + Z \leq 10$	48	48
Maximize $S = 8W + 4X + 12Y + 6Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + Z \leq 10$	52	52
Maximize $S = 8W + 4X + 13Y + 6Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + Z \leq 10$	52.67	52.6667
Maximize $S = 8W + 4X + 13Y + 8Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $10W + 8X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + Z \leq 10$	67.3333	67.3333
Maximize $S = 8W + 4X + 13Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$	74.67	74.6667

$10W + 8X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + Z \leq 10$		
Maximize $S = 4W + 6X + 13Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + 4Z \leq 10$	32	32
Maximize $S = 4W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + 4Z \leq 10$	35	35
Maximize $S = 10W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 40$ $2W + 4X + 4Y + 4Z \leq 10$	50	50
Maximize $S = 10W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 40$	126.67	126.667
Maximize $S = 9W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 40$	123.333	123.333
Maximize $S = 9W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 30$ $2W + 4X + 5Y + 5Z \leq 20$	90	90
Maximize $S = 9W + 6X + 14Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 10$ $2W + 4X + 5Y + 5Z \leq 20$	46.67	46.6667
Maximize $S = 20W + 18X + 3Y + 10Z$ subject to $15W + 10X + 3.75Y + 3Z \leq 20$ $5W + 8X + 4Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 20$ $10W + 10X + 10Y + 10Z \geq 10$	53.61	53.617
Maximize $S = 9W + 8X + 5Y + 10Z$ subject to $15W + 10X + 3.75Y + 3Z \leq 20$ $5W + 8X + 4Y + 5Z \leq 40$ $7W + X + 2Y + 5Z \leq 20$ $10W + 10X + 10Y + 10Z \geq 10$	45.10	45.1064
Maximize $S = 9W + 6X + 15Y + 9Z$ subject to $3W + 6X + 3Y + 3Z \leq 10$ $2W + 4X + 5Y + 5Z \leq 20$	50	50
Maximize $S = 3W + 6X + 16Y + 10Z$ subject to $3W + 6X + 3Y + 3Z \leq 10$ $2W + 4X + 5Y + 5Z \leq 20$	53.3333	53.3333
Maximize $S = 3W + 7X + 16Y + 15Z$ subject to	64	64

$3W + 6X + 3Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$		
Maximize $S = 3W + 7X + 16Y + 15Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	61.5	61.5
Maximize $S = 3W + 7X + 15Y + 15Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	60	60
Maximize $S = 3W + 7X + 14Y + Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	42	42
Maximize $S = 3W + 7X + 14Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	46	46
Maximize $S = 3W + 7X + 13Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	44.5	44.5
Maximize $S = 3W + 7X + 12Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	43	43
Maximize $S = 3W + 7X + 11Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	41.5	41.5
Maximize $S = 3W + 7X + 10Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	40	40
Maximize $S = 3W + 7X + 9Y + 9Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	36	36
Maximize $S = 3W + 7X + 9Y + 8Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	33.5	33.5
Maximize $S = 3W + 7X + 9Y + 7Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	31	31
Maximize $S = 3W + 7X + 9Y + 6Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	28.5	28.5
Maximize $S = 3W + 7X + 9Y + 5Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	27	27
Maximize $S = 3W + 7X + 8Y + 9Z$ subject to $3W + 6X + 5Y + 3Z \leq 15$ $2W + 4X + 5Y + 5Z \leq 20$	36	36

Maximize $S = 3W + 9X + 9Y + 9Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	37	37
Maximize $S = 3W + 9X + 12Y + 9Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	39	39
Maximize $S = 3W + 9X + 12Y + 10Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	42	42
Maximize $S = 3W + 9X + 12Y + 11Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	45	45
Maximize $S = 3W + 9X + 12Y + 12Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	48	48
Maximize $S = 3W + 9X + 12Y + 13Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	52	52
Maximize $S = 3W + 9X + 12Y + 14Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	56	56
Maximize $S = 3W + 9X + 12Y + 15Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	60	60
Maximize $S = 3W + 9X + 12Y + 16Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	64	64
Maximize $S = 3W + 9X + 12Y + 17Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	68	68
Maximize $S = 3W + 9X + 12Y + 18Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	72	72
Maximize $S = 3W + 9X + 12Y + 19Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	76	76
Maximize $S = 3W + 9X + 12Y + 20Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	80	80
Maximize $S = 10W + X + Y + 20Z$ subject to $3W + 6X + 5Y + 3Z \leq 14$ $2W + 4X + 5Y + 5Z \leq 20$	82.2222	82.2222
Maximize $S = 10W + X + Y + 20Z$ subject to	88.89	88.8889

$3W + 2X + 7Y + 3Z \leq 20$ $2W + 4X + 5Y + 5Z \leq 20$		
Maximize $S = 10W + X + Y + 20Z$ subject to $3W + 2X + 7Y + 3Z \leq 20$ $2W + 4X + 5Y + 5Z \leq 30$	122.222	122.222
Maximize $S = W + X + Y + 2Z$ subject to $3W + 2X + 7Y + 3Z \leq 20$ $2W + 4X + 5Y + 5Z \leq 30$	12.2222	12.2222
Maximize $S = W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	20	20
Maximize $S = 2W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	40	40
Maximize $S = 3W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	60	60
Maximize $S = 4W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	80	80
Maximize $S = 5W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	100	100
Maximize $S = 5.5W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $W + 4X + 5Y + 5Z \leq 30$	110	110
Maximize $S = 5.5W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	41.25	41.25
Maximize $S = 6W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	45	45
Maximize $S = 7W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	52.5	52.5
Maximize $S = 8W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	60	60
Maximize $S = 8.5W + X + Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	63.75	63.75
Maximize $S = 8.7W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	65.25	65.25

Maximize $S = 9W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	67.5	67.5
Maximize $S = 9.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	71.25	71.25
Maximize $S = 9.7W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	72.75	72.75
Maximize $S = 10W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 20$ $4W + 4X + 5Y + 5Z \leq 30$	75	75
Maximize $S = 10W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	62.5	62.5
Maximize $S = W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	14.3	14.2857
Maximize $S = 2W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	17.86	17.8571
Maximize $S = 2.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	19.64	19.6429
Maximize $S = 3W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	21.43	21.4286
Maximize $S = 3.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	23.21	23.2143
Maximize $S = 4W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	25	25
Maximize $S = 4.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	28.125	28.125
Maximize $S = 5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	31.25	31.25
Maximize $S = 5.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	34.375	34.375
Maximize $S = 6W + X + 5Y + 2Z$ subject to	37.5	37.5

$W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$		
Maximize $S = 6.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 25$ $4W + 4X + 5Y + 5Z \leq 25$	40.625	40.625
Maximize $S = 6.5W + X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $4W + 4X + 5Y + 5Z \leq 35$	56.87	56.875
Maximize $S = 7W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $4W + 4X + 5Y + 5Z \leq 35$	61.25	61.25
Maximize $S = 7.5W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	52.5	52.5
Maximize $S = 7.7W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	53.9	53.9
Maximize $S = 8W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	56	56
Maximize $S = 8.5W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	59.5	59.5
Maximize $S = 8.7W + 4X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	60.9	60.9
Maximize $S = 9W + 15X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	131.25	131.25
Maximize $S = 9W + 10X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	87.5	87.5
Maximize $S = 11W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	77	77
Maximize $S = 11.5W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	80.5	80.5
Maximize $S = 11.7W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	81.9	81.9
Maximize $S = 12W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	84	84

Maximize $S = 13W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	91	91
Maximize $S = 13.5W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	94.5	94.5
Maximize $S = 14W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 28$ $5W + 4X + 7Y + 5Z \leq 35$	98	98
Maximize $S = 14W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	42	42
Maximize $S = 5W + 5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	18.75	18.75
Maximize $S = 5W + 6X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	22.5	22.5
Maximize $S = 5W + 6.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	24.375	24.375
Maximize $S = 5W + 7X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	26.25	26.25
Maximize $S = 5W + 7.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	28.125	28.125
Maximize $S = 5W + 8X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	30	30
Maximize $S = 5W + 9X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	33.75	33.75
Maximize $S = 5W + 9.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 18$ $5W + 4X + 7Y + 5Z \leq 15$	35.625	35.625
Maximize $S = 5W + 10X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	37.5	37.5
Maximize $S = 5W + 10.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	39.375	39.375
Maximize $S = 5W + 11X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$	41.25	41.25

$5W + 4X + 7Y + 5Z \leq 15$		
Maximize $S = 5W + 11.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	43.125	43.125
Maximize $S = 5W + 11.7X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	43.875	43.875
Maximize $S = 5W + 12X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	45	45
Maximize $S = 5W + 12.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	46.875	46.875
Maximize $S = 5W + 12.7X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	47.625	47.625
Maximize $S = 5W + 13X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	48.75	48.75
Maximize $S = 5W + 13.5X + 5Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	50.625	50.625
Maximize $S = 5W + 14X + 10Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 15$ $5W + 4X + 7Y + 5Z \leq 15$	52.5	52.5
Maximize $S = 5W + 15X + 14Y + 2Z$ subject to $W + 2X + 10Y + 3Z \leq 10$ $5W + 4X + 7Y + 5Z \leq 10$	37.5	37.5

4.2 Result Conclusion

Comparing the aforementioned results of both methods, we can safely claim that our algorithm is successful for solving Linear program models with four decision variables.

Chapter 5

Standards, Ethics, Impacts & Challenges.

5.1 Standards

As works of academic nature, high standards were maintained to ensure that our methodology and proofs were up to standards.

5.2 Ethics

As our work is of an academic nature, there isn't any ethical concerns about its applications or implementations in any nature of work, and if there are any future works that make use of our methodology and requires ethical bounds within which to function, it is not our jurisdiction to oversee these rules.

5.3 Impacts

Our method of solving linear optimization problems holds huge possibility of having an impact on the business and academic fields of operational research. It unlocks the possibility of instinctively understanding the operations that are happening while finding the optimized solution as well as toning the down the number of steps that need to be executed in order to find the optimum solution to a finite and static number.

whereas, based on the number of decision variables or constant coefficient variety the algebraic method can vary on the number of instructions it needs to execute in order to come up with the solution

5.4 Challenges

Linear programming operations has been in through usage for a long time now, and new methods such as ours are still in testing phase in terms of probable spooky conditions, In order to reliably use our method practically, it has to be thoroughly tested for massive amount of cases and proved reliable.

Chapter 6

Constraints

6.1 Design Constraints

The overall structure of the proposed architecture can be implemented as a four-dimensional graph projected on a two-dimensional plane, as we are using pseudo-special dimensions. Finding a graph engine that supports such rich set of operations was quite impossible for us, so in the end we had to make our own.

6.2 Budget Constraints

There was thankfully, no budget constraint as our research field wasn't projecting existing academic frameworks on practical problems but rather, purely academic.

6.3 Summary

Due to the notion that higher dimensional graph engines would be quite impossible and not to mention improbable to make visualization and projection of higher dimensional objects was quite a problem.

Chapter 7

Schedules, Tasks & Milestones

7.1 Timeline

The overall timeline of the thesis work can be divided into three segments based on the three semesters worth of work execution procedure, implemented by our supervisor.

The first-semester work process contains the planning and reviewing of the related works of the thesis work. The second-semester work process contains collaborative work of prototype designing and analysis of the prototype. In the third semester, we implemented and tested the overall model, along with reporting the overall workflow.

7.2 Gantt Chart

Figure 7.1 contains the Gantt chart describing the work execution process of the thesis works. The overall execution of the thesis work is three semesters long, where each semester is twelve weeks long.

Chapter 8

Conclusion

In this report a novel graphical method of working with linear optimization problems has been proposed, to test the limitation put upon it by the evaluation of the decision variables and to further make ease of use, of such work, not just computationally but also comprehensively. Although, this method has not yet inherited mathematical finesse, it provides a way forward, where linear optimization problems can be solved using human intuition and instinct and not by just using cryptic cumbersome calculations, which in first sight may as well seem confusing to its users.

The proposed method, theoretically speaking has the capability to solve for any number of decision variables and it does so, to a certain extent, but as a method relying on human intuition it falls short when working with massive numbers of decision variables, which is not per say a faultiness within the method but rather another supposed limitation on human visualization. Our future work will be focused on trying to eliminate this limitation altogether and not just present an intuitive method for solving linear optimization problems, but a reliable and operationally superior one as well.

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